Problem 10: The Palindrome Train Code

TASK

Each train has a code stored as a singly linked list of digits. Codes that read the same from both ends are reserved for special trains. The check must use constant extra space, and the list must be left exactly as it was.

Implement `boolean isPalindrome(Node head)` in O(n) time and O(1) extra space, restoring the list before returning. Use your own node class (for example `class Node { int data; Node next; }`); do not use `java.util.LinkedList` or other collections for the list itself.

CONSTRAINTS

- 1 <= length of list <= 10^5
- O(1) additional space; no array or stack copies
- The list must be unchanged after the call

EXAMPLES

Sample 1

```
code = 1 -> 2 -> 3 -> 2 -> 1

Expected Output: true
```

Sample 2

```
code = 1 -> 2 -> 3 -> 2

Expected Output: false
```

Sample 3

```
code = 1 -> 2 -> 2 -> 1

Expected Output: true (list afterwards: 1 -> 2 -> 2 -> 1)
```

COMPLEXITY REQUIREMENTS

State the time and space complexity, and compare them with copying the values into an array.

HINT

> *Find the middle, reverse the second half, compare the two halves, and then reverse the second half again to restore the list.*